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Case Report

Polyacrylamide hydrogel pulmonary embolism—A fatal consequence of an illegal cosmetic vaginal tightening procedure: A case report

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ABSTRACT

Vaginal tightening is a kind of esthetic surgery aimed at enhancing sexual satisfaction during intercourse. Although the injective vaginal tightening procedure is informal, there are already some reports of its application. But pulmonary embolism is a really rare therapeutic complication of this procedure. We report a case of death due to the non-thrombotic pulmonary embolism as a consequence of illegal cosmetic vaginal-tightening procedure using polyacrylamide hydrogel. A 34-year-old woman was hospitalized with paroxysmal abdominal cramps and diarrhea as initial symptoms, while she concealed the genital cosmetic surgery history. Respiratory distress presented only 1.5 h before she died. The result of autopsy revealed the cause of death as pulmonary embolism due to the hydrogel which was injected into her vaginal wall. The emboli were confirmed as polyacrylamide hydrogel by Alcian-blue stain and the Fourier transform infrared scanning. It is suggested that pulmonary embolism should not be discarded in the expertise of deaths following cosmetic implant surgeries. It broadens our understanding about death associated with esthetic genital procedures and informs clinicians and medical examiners of the potential death of this type. And detailed investigations of previous medical and surgical history will always play a critical role in the certification of cause of death.

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1. Introduction

Female esthetic genital surgery, a relatively new category in the field of cosmetic and plastic surgery, has become a fashionable issue nowadays. This type of operation is usually performed by gynecologists in order to repair the genital size, appearance and function after obstetrical delivery, perineorrhaphy, intersex and transsexual surgical procedure, before the appearance of plastic procedures [1]. Such procedures promised sexual enhancement and cosmetic improvement for both women and their sexual partners. Female esthetic genital surgery mainly includes vaginoplasty/vaginal rejuvenation/vaginal tightening, perineoplasty,

labia minora labiaplasty, clitoral hood reduction, labia majora augmentation, and G-spot amplification [2]. The American College of Obstetricians and Gynecologists' Committee has warned that absence of data supporting the safety and efficacy of these procedures makes their effects untenable [3], but this cannot abate women's enthusiasm for trying to enhance function, appearance, and self-confidence of their bodies.

Vaginal tightening surgery appeared in the mid-1950s and no standardization of the procedures exists to date. Usually, the gynecologists or cosmetic surgeons perform anterior colporrhaphy, high-posterior colporrhaphy, excision of the lateral vaginal mucosa, or a combination of the above to reduce the caliber of the vaginal canal for the purpose of tightening the relatively lax vagina [2,4,5]. And these techniques are derived from the classic gynecologic surgical treatments applied to pelvic floor defects, such as cystocele, rectocele and enterocele [4]. Despite the techniques utilized, filler injection remains experimental, and there are already some reports of their application. The implants mentioned in the literatures include autologous fat, the bulking

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agent, hyaluronic acid, and silicone [2,6,7]. In clinical and forensic practice, death associated with non-thrombotic pulmonary embolism (NTPE) caused by injection for vaginal tightening is rare. There is no report about this type of death in existing literatures. Only Park et al. [6] reported a case of NTPE due to hyaluronic acid injection for G-spot amplification in which the patient was alive.

Here, we report the case of a woman who died of NTPE due to illegal cosmetic vaginal tightening procedure using polyacrylamide hydrogel (PAAG).

2. Case report

A 34-year-old woman was admitted in hospital for paroxysmal abdominal cramps and watery diarrhea for several hours, as well as tenesmus, nausea and vomiting. Physical examination showed a weak patient with pale skin and mucosa, tachycardia and local tenderness below the xiphoid region. The admitting diagnosis was shock of undetermined origin and suspected bromatoxism. She was treated with intravenous ofloxacin and metronidazole; however, she died of cardiopulmonary arrest 6 h after admission. Medical history profile showed that the deceased underwent vaginal tightening surgery by injecting PAAG (Amazing Gel) into vaginal wall 1 day earlier.

Forensic necropsy was performed 24 h after her death. External examination showed a well-nourished adult woman without any signs of mechanical injury and mechanical asphyxia. Blood clots were seen in the perineum and vagina. Internal examination revealed edematous lungs with diffuse petechiae. Multiple hemorrhages were observed in several locations, including serosa, mucosa of small intestine, and pancreas. Splinter hemorrhages were found on the serosa and corresponding submucosa of the fundus vesicae. On incision of the dextral vaginal wall, several hematomas and bloody gelatinous material were found.

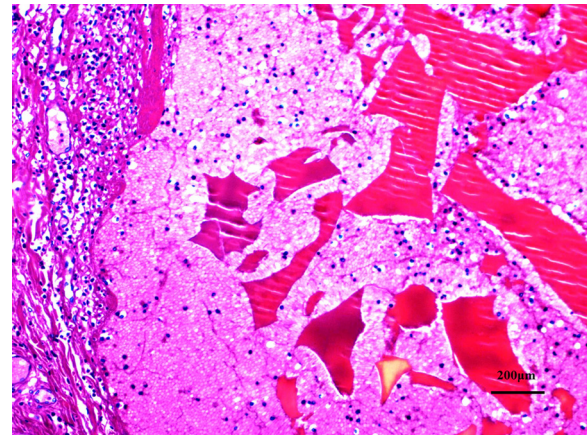


Fig. 1. Hematomas containing jelly-like amorphous substance formed in vaginal wall (HE stain, $\times 100$).

Significant hematomas were found in the vaginal wall, as well as a jelly-like amorphous substance (Fig. 1), fibrinous exudate, and prominent infiltration of inflammatory cells. Furthermore, blood vessel in submucosa and muscular layer of vagina were filled with a basophilic amorphous material. Interestingly, the basophilic material was also found in the vessels of lungs with severe pulmonary edema and focal pulmonary hemorrhages. No such material emboli were found in other organs. The samples of vaginal wall and pulmonary emboli were processed with Alcian-blue staining, and the gelatinous material were positive (Fig. 2).

Infrared spectra were obtained by FTIR (Thermo Scientific Nicolet TM7500-II, USA) equipped with continuum infrared microscope under the guiding of the unstained section. Infrared spectra were recorded in the range of $4000\text{--}900\text{ cm}^{-1}$ at a

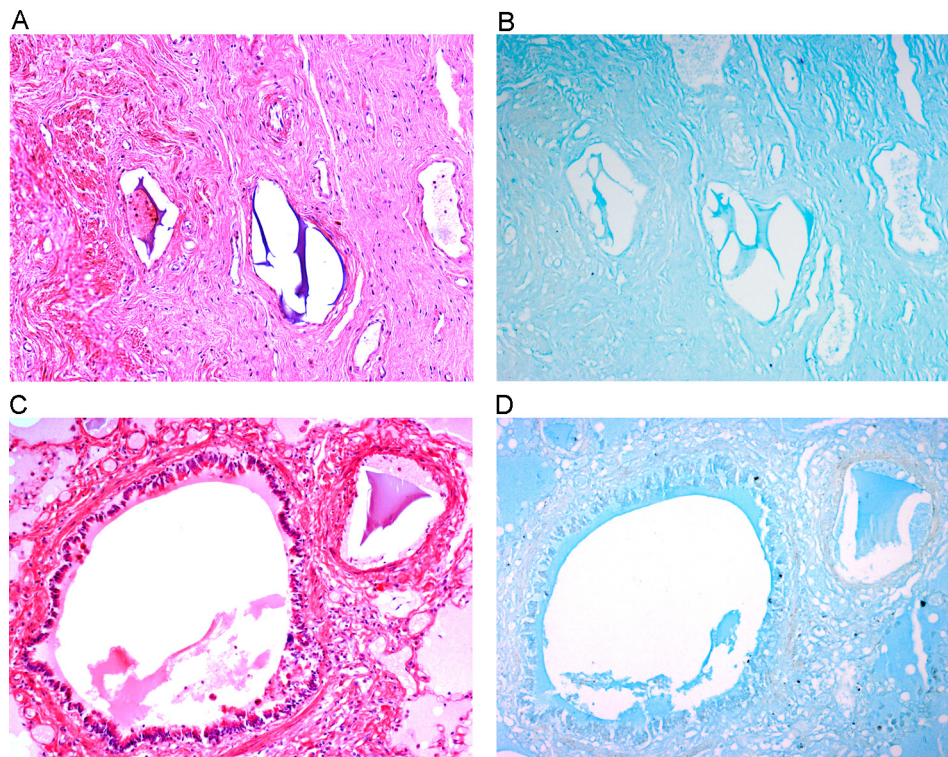


Fig. 2. (A) Basophilic gel emboli in blood vessel in submucosa and muscular layer of vagina (HE stain, $\times 200$); (B) the emboli in vaginal wall stained positive (Alcian blue stain, $\times 200$); (C) emboli in pulmonary vessels and alveolar capillaries (HE stain, $\times 200$); (D) the gelatinous material in pulmonary arterioles was positive (Alcian blue stain, $\times 200$).

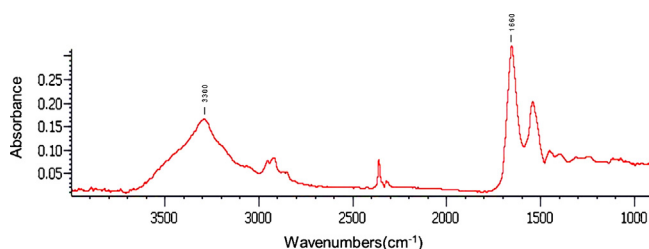


Fig. 3. Infrared spectrum of the gel emboli located in pulmonary arteriole confirmed the presence of PAAG at frequencies 3300 cm^{-1} and 1660 cm^{-1} .

resolution of 8 cm^{-1} , with an aperture setting of $60\text{ }\mu\text{m} \times 60\text{ }\mu\text{m}$. The spectra had excellent signal intensities and signal-to-noise ratios (Fig. 3). The component of the gelatinous material was confirmed to be PAAG in infrared spectra, with band presenting around the frequency of 3300 cm^{-1} and 1660 cm^{-1} [8]. In FTIR mapping (FTIRM), the gelatinous material emboli located in the vaginal wall and pulmonary arteriole showed increased absorption intensity at these characterized peaks (Fig. 4).

The cause of death was determined as pulmonary embolism due to PAAG vaginal injection for vaginal-tightening purpose.

3. Discussion

Polyacrylamide is a polymer formed by the polymerization of acrylamide monomers, and is believed to be a nontoxic and non-degradable material. PAAG is composed of 2.5% polyacrylamide and 97.5% non-pyrogenic water [9]. It has been used as a medical material and a biocompatible permanent orthopedic filler in Ukraine since 1980 [10]. For its good stability and resistance to degeneration, it was supposed to be nontoxic and definitely efficient. It was approved by the European Committee in 2001 for application in facial contouring and breast and other soft tissue augmentation. According to different manufacturers, there are a variety of polyacrylamide hydrogel products such as Aquamid, Amazing Gel, Interfall, Outline, Formacryl, Bioformacryl, Bio-Alcamid, and Argiform [11]. In China, a commercial product of

PAAG was imported from Ukraine and it was approved by the State Drug Administration in China in 1997 [8,12]. The most marketed products in China were Aquamid (Contura International, Denmark) and Amazing Gel (FuHua Ltd., China). With the wide application of PAAG, increase in adverse reactions was observed. The complications associated with PAAG injection includes focal lumps, inflammation, infection, asymmetry, hematomas, irregularity of the injection site, keloid formation, gel migration, pain in cold weather and feeling of expansion in a warm environment, skin necrosis and hyperpigmentation [9,11,13,14]. The Chinese State Food and Drug Administration (SFDA) received 183 reports of adverse effects from the use of PAAG from 2002 to 2005, and most of these reports involved breast implant patients who suffered infections and disfigurement [12]. Soon after, the marketing of PAAG for any cosmetic purpose was prohibited by SFDA in 2006. Despite the fact that PAAG used for vaginal tightening has not been recommended in formal medical institutions, illicit procedures are performed in many clinics and beauty parlors. It is believed that approximately 300,000 patients had plastic surgery using the illicit PAAG products before 2006 [8]. Besides this case, we have reasons to believe that there is still illegitimate use of PAAG in China. The low cost to patients remains their paramount motivation.

Amin et al. [9] considered that most PAAG injection-related acute complications are usually minor and transient; whereas, embolization should be classified as a delayed complication. However, there are more literatures supporting that embolism due to filler injection usually arises in the early post-operative period and often leads to some unfavorable outcomes, even death [6,7,15,16]. The mechanism of NTPE arising from cosmetic filler implantation may occur in a number of ways, such as high-pressure injection, large injected volume, massage or trauma at the injection site after the procedure, and direct intravenous injection [7,16]. The injective layer of PAAG is initially suggested to be the subcutaneous fat, the muscle, or the supra-periosteal plane [10]. However, to avoid migration of the implant, some researchers considered that PAAG should not be injected into muscular tissue or subcutaneous areas with active movement [17]. As extensive venous plexus immediately surrounds the vagina while movement of the muscle group around it is relatively frequent, vaginal

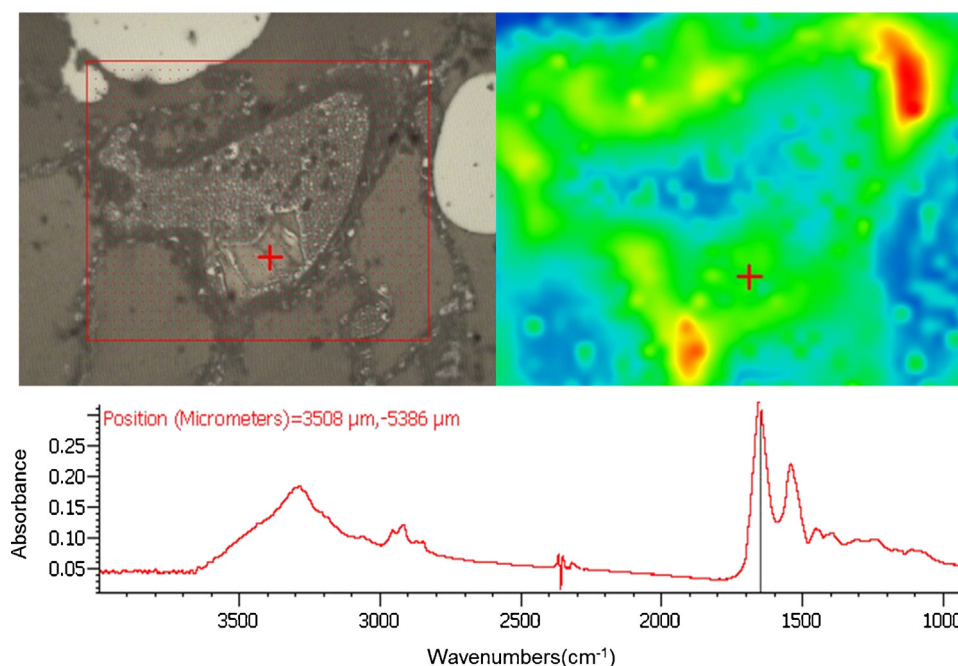


Fig. 4. The gelatinous material emboli in pulmonary arteriole show increased absorption intensity at 1660 cm^{-1} in FTIRM. (The red plus indicates the gel emboli.)

cosmetic procedures using injectable materials can bring out potential complications including NTPE. In this case, the interval between PAAG injection performed and the occurrence of death was nearly 1 day, which suggested that the hydrogel was not injected directly into the vessel. According to the anatomical features of vagina and the physical properties of PAAG, we considered that migration of PAAG caused by the activities of pelvic floor muscle played a critical role in the mechanism of NTPE. The migratory hydrogel may have traveled along the blood vessels, through plexus venosus vaginalis, internal iliac vein, postcava and right in sequence, and finally entered the pulmonary vasculature. The absence of a guide for the dosage of PAAG to be used in the injection for vaginal tightening may also have contributed to NTPE.

As the cardinal symptoms and signs of NTPE are not specific, obtaining detailed history plays a critical role in the diagnosis of NTPE. Sometimes, specific computed tomography (CT) imaging finding and some laboratory results can also reveal the possibility of NTPE [18]. However, because of the lack of positive signs or limited medical condition, not all patients with NTPE would receive CT examination after hospitalization. Also sometimes, the medical examiner cannot obtain the complete medical records when the autopsy is performed. Based on the reasons given above, it is difficult for the medical examiners to make a diagnosis of the NTPE. Major manifestations of the woman in our case were only paroxysmal abdominal cramps and watery diarrhea. Respiratory failure presented only 1.5 h before her death. She did not receive any medical imaging examination before her death.

In this case, we encountered another problem with watery diarrhea being the only positive sign of our deceased. Whether the diarrhea was caused by the cosmetic procedure or it was just an independent condition is an issue worth solving. A case of acute ischemic colitis secondary to air embolism was reported by Payor and Tucci [19] have provided four suspected mechanisms of how the formed gas bubbles enter the systemic arterial circulation. Some reports of systematic emboli due to silicone exist, too. Although there was obvious inflammatory reaction in the intestinal wall, gel embolus was not observed in the vessels in the intestinal wall and mesenterium. Hence, we considered that there was no sufficient evidence that the diarrhea was caused by the PAAG injection directly.

In addition, even if the polyacrylamide is nontoxic, several studies showed that exposure of humans and laboratory animals to monomeric forms of acrylamide causes neurotoxicity, genotoxicity, carcinogenicity, mutagenicity and reproductive toxicity [20–22]. The earliest literature that can be retrieved about the toxicology of acrylamide was by Kuperman [23]. He found that repeated sublethal doses gave rise to a chronic and reversible intoxication, characterized by ataxia and postural and ataxic tremors. Severe tonic-clonic convulsions and other signs of a diffuse central excitation were produced by lethal doses. McCollister et al. [21] revealed that acrylamide was toxic whether absorbed cutaneously, orally, or parenterally, and all species showed involvement of the nervous system. Meanwhile, residual acrylamide monomer is likely an impurity in most polyacrylamide preparations [20]. Due to the above reasons, whether the toxicity of residual acrylamide contributed to the death is an inevitable problem in this case. The main signs and symptoms of acrylamide intoxication are numbness of the limbs; ataxia and skeletal muscle weakness; increased sweating of the hands, with erythema and peeling of the palms; and tiredness and lethargy [24,25]. The deceased in our case did not show any symptoms, especially neurological signs, indicating the presence of acrylamide poisoning. Besides, the duration from mobility to mortality was just about 24 h, which does not conform with the characteristics of acrylamide toxicity which is usually subchronic and cumulative

[25,26]. Thus, a combination of clinical manifestations could preclude intoxication of residual acrylamide in the implant.

Death due to the NTPE associated with injective vaginal tightening procedure is rare in both clinical and forensic practice. Low specificity of the clinical manifestation makes the clinical diagnosis difficult. Diarrhea, hematuria and bloody stools can be prompted by the vaginal wall injection. Medical history, especially the cosmetic operation, should be investigated exhaustively. In addition, detailed necropsy and histopathology examinations should be performed. And PAAG is positive with Alcian blue stain and it is not birefringent under polarizing microscopy [27]. Fourier transform infrared (FTIR) microspectroscopy can also be used in the identification of PAAG [8].

Conflict of interest

None.

Acknowledgments

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